SAILOR: Systematic Tool Support for Systematic Literature Reviews

Jan Keim
Karlsruhe Institute of Technology (KIT)
Karlsruhe, Germany
jan.keim@kit.edu

***Abstract*—Systematic literature reviews are a common research method in software engineering. They follow a protocol so that other researchers can trace and repeat each decision. However, the review data is hard to keep traceable. Reviewers record screening decisions and extracted values in spreadsheets, shared documents, or hosted services. There, the link between an extracted value and its evidence in the paper can get lost. Also, it is hard to reconstruct who decided what and when. To tackle this, we present SAILOR, an open-source desktop tool that supports screening, data extraction, and consolidation of a review in one project. The project is a set of plain JSON files that a schema describes and that git versions. Several reviewers work independently in their own seats, and a consolidator resolves their disagreements with agreement statistics. When reviewers synchronize through git, the tool merges concurrent changes field by field. Optional large language model support proposes answers with evidence, and a human reviews every proposal. We describe the tool from the reviewer's view and explain its main architecture decisions, including the move from Electron to Tauri. We also report how we keep the tool itself traceable. We documented 389 requirements and linked them to code, tests, and documentation. Of these requirements, 93% link to implementing source code and 92% link to at least one automated test. The test suites hold 2834 unit tests, 56 integration tests, and 104 Rust tests. We release SAILOR 2.0.0 under the GPL-3.0 license and archive it on Zenodo.**

***Index Terms*—Systematic Literature Review, Tool Support, Data Extraction, Requirements Traceability, Large Language Models**

I. INTRODUCTION

systematic literature reviews (SLRs) are an established research method in software engineering (SE). Kitchenham and Charters [1] adapted the method from medicine, and later guidelines added systematic mapping studies and snowballing [2], [3]. The core promise of an SLR is that other researchers can follow each step. Therefore, reviewers define a protocol up front, select papers with explicit criteria, and extract data with a fixed form. Reporting guidelines such as Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [4] then ask authors to disclose how many papers they found, screened, and excluded, and why.

However, the data that an SLR produces is hard to keep traceable. Reviewers often record screening decisions and extracted values in spreadsheets, shared documents, or hosted services. In such a setup, the link between an extracted value and the passage in the paper that supports it is often only in the reviewer's head. Also, two or more reviewers usually extract the same papers independently, and someone has to compare and consolidate their answers. With a shared spreadsheet, it is hard to see who changed which cell and why. Hassler et al. [5] collected barriers to the SLR process in a community workshop. Recently, large language models (LLMs) promise to take over parts of screening and extraction. However, studies in SE and other fields find that their output still needs human control [6], [7].

Several tools support parts of the process. Mapping studies and evaluations cover many of them [8]–[10]. Many of these tools are hosted web applications that keep the review data on a server, for example, Covidence [11], Rayyan [12], and CADIMA [13]. Other tools focus on one activity, such as screening with active learning in ASReview [14]. To the best of our knowledge, no tool combines local, plain-file review data with a version history, schema-driven extraction with evidence links, independent reviewers with a consolidation step, and LLM proposals that a human must accept.

In this paper, we present SAILOR, an open-source desktop tool that supports SLRs in a systematic way. The name contains the letters S, L, and R of "systematic literature review". The main idea is to treat review data like source code. A review is a project of plain JSON files. A schema describes what reviewers extract, and the tool validates every answer against it. Each reviewer writes into their own files, and git records every change. When reviewers pull each other's work, the tool merges their files field by field. Optional LLM assistance proposes answers with quotes from the paper, but the reviewer decides which proposal to apply, and the project records that an LLM was involved.

We use the example project that ships with SAILOR as a running example. The project is a small review on "Code Search and Program Repair". A screening project holds eight candidate papers. Five of them are included and carried into an extraction project with a nested schema. Figure 1 shows this extraction project in the main view of SAILOR.

In summary, we make the following contributions:

- We present SAILOR, a desktop tool that supports screening, data extraction, and consolidation of an SLR with several reviewers.
- We describe the main architecture decisions behind SAILOR and their reasons, including local-first storage, git as collaboration layer, the move from Electron to Tauri, and the control of LLM assistance.

- We report how we keep the tool traceable with 389 requirements that we linked to source code, tests, and documentation.
- We release SAILOR 2.0.0 as open source under the GPL-3.0 license on GitHub and Zenodo [15].

The remainder of the paper is structured as follows: In Section II, we discuss related work. In Section III, we explain which SLR activities SAILOR supports and which design principles guide it. Section IV walks through the tool from the reviewer's view. Section V describes the main architecture decisions. In Section VI, we present the requirements and their trace links. We discuss limitations in Section VII and conclude in Section VIII.

II. Related Work

In this section, we give an overview of related work on tool support for SLRs. We first look at the methodology and at studies about SLR tools. Then, we discuss existing tools and, lastly, the use of LLMs in SLRs.

A. Methodology and Studies on Tool Support

The SLR method in SE goes back to the guidelines by Kitchenham and Charters [1] and the book by Kitchenham et al. [16]. Petersen et al. [2] describe systematic mapping studies, and Wohlin [3] describes snowballing as a search strategy. PRISMA 2020 [4] defines what a systematic review reports, including a flow diagram of found, screened, and included studies. These guidelines define the activities that SAILOR supports. SAILOR does not add new methodology. Instead, it records the data that these guidelines ask for.

Several studies look at tools for SLRs. Marshall and Brereton [8] map the tools that SE researchers proposed, and Marshall et al. [9] compare the features of SLR tools in SE. Hassler et al. [5] collected barriers to the SLR process in a community workshop. Al-Zubidy and Carver [17] derived requirements for tools that support the search activity. Felizardo and Carver [18] give an overview of how SLR activities can be automated, and van Dinter et al. [19] review automation approaches in an SLR of their own. Most of the studies that van Dinter et al. found automate the selection of primary studies. Outside SE, Harrison et al. [10] evaluate tools for title and abstract screening in healthcare. These studies show that tool support exists for many activities. Our work differs in its goal. We do not survey or compare tools but describe the design of one tool and how we keep it traceable.

B. Tools for Systematic Literature Reviews

A large group of tools are hosted web applications. Covidence [11] is a web platform that covers screening, full-text review, risk of bias, and data extraction. Rayyan [12] is a web and mobile app that supports screening with semi-automated suggestions. CADIMA [13] is an open-access online tool by the Julius Kühn-Institut for the conduct and reporting of systematic reviews and maps. Parsifal [20] is an online tool that supports researchers in planning, conducting, and reporting SLRs. In all four tools, the review data lives on a server, and collaboration happens through the shared online project. SAILOR takes the opposite route. The review data stays in local files, and reviewers exchange it through git, a hosting service of their choice, or not at all. Therefore, reviewers can inspect, archive, and publish the raw data with standard tools.

Some tools come from SE research. StArt [21], [22] supports the whole systematic review process, and Hernandes et al. [21] evaluated it with GQM and TAM. SLR-Tool [23] is a free tool that follows the process by Kitchenham and adds text mining and export to BibTeX and RIS. Both tools target the same community as SAILOR. The publications on these tools do not cover independent extraction by several reviewers with a consolidation step, and they do not cover collaboration through version control.

ASReview [14] is an open-source framework that uses active learning to rank the records that a reviewer should screen next. Like SAILOR, ASReview can run locally. However, ASReview focuses on screening, while SAILOR focuses on extraction and consolidation and offers screening as a preceding step. SAILOR does not rank records with active learning. The two tools can complement each other, because SAILOR imports BibTeX, RIS, and CSL-JSON files.

C. Large Language Models in Systematic Literature Reviews

Recently, many studies evaluate LLMs for single SLR activities. For selection, Felizardo et al. [6] evaluate the accuracy of ChatGPT on SLRs in SE, and Syriani et al. [24] use ChatGPT to screen articles for systematic reviews. Felizardo et al. report 75.3% and 86.1% accuracy in the selection activity of two SLRs. They advise against fully outsourcing the selection to ChatGPT, but see it as a support tool. For extraction, Felizardo et al. [25] and Gartlehner et al. [26] report proof-of-concept studies. Khraisha et al. [7] test GPT-4 for screening and extraction across languages. After correcting for chance agreement and dataset imbalance, they find only moderate performance for data extraction. Other work uses ChatGPT to formulate Boolean search queries [27] or across several stages of a review [28]. These studies evaluate the models. They do not address how a tool should present model output so that the reviewer stays in control and the use of the model stays visible in the report. SAILOR addresses this second question. Every proposal is a suggestion that a reviewer must apply, the model can act as an additional reviewer whose answers go through the same consolidation, and the project records each use of the model.

Overall, existing tools either keep the review data on a server or focus on a single activity. To the best of our knowledge, no tool yet combines local, versioned review data, schema-driven extraction with evidence links, consolidation of independent reviewers, and controlled LLM assistance.

III. Supporting SLRs Systematically

In this section, we explain which SLR activities SAILOR supports and which principles guide its design. We then introduce the running example in more detail.

TABLE I
SLR PHASES AND ACTIVITIES [1] AND THEIR SUPPORT IN SAILOR

Activity	Support in SAILOR
Planning	
Protocol	Research questions, search strings, databases, search date, and notes in the project
Extraction form	Schema of nested fields with types, options, and visibility conditions
Conducting	
Search	Not supported, import of BibTeX, RIS, CSL-JSON, and PDF folders
Duplicates	Classification into new, certain, and probable duplicates with a human decision
Screening	Include, exclude with reason, or leave undecided per reviewer
Extraction	Schema-driven form next to the PDF with highlights and evidence links
Consolidation	Comparison of reviewers with agreement statistics and adoption of answers
Reporting	
Selection counts	Screening summary with counts per exclusion reason
Exports	Disagreements as text, marks as standard PDF annotations
LLM use	Durable record of each applied LLM proposal per paper
Synthesis	Not supported, extracted data as plain JSON

A. Covered Activities

Kitchenham and Charters [1] split an SLR into three phases: planning, conducting, and reporting. Table I maps these phases to the support in SAILOR. The focus lies on the conducting phase, where several reviewers handle the same papers and most of the review data comes up. SAILOR does not run the database search itself. Reviewers search in digital libraries as before and import the exported references. SAILOR also does not synthesize the results. Synthesis depends on the research questions, and the extracted data is plain JSON that reviewers can analyze with their own scripts.

B. Design Principles

We built SAILOR on five principles. Each principle follows from the goal that other researchers should be able to trace the review data.

a) Local, plain files: A project is a `project.json` file with the metadata, the schema, and the papers. A sibling `annotations` folder holds one file per paper and reviewer. Kleppmann et al. [29] argue that users should own their data and keep working without a server. For an SLR, local files have a further benefit. Reviewers can publish the raw review data with the paper, and readers do not need SAILOR to read it. To keep diffs small, SAILOR writes files in a deterministic order and does not write anything when a project is only opened. It also keeps keys in the files that it does not know.

b) The schema drives extraction: Reviewers define the extraction form as a schema. A schema consists of nodes. A node is either a field of type string, number, boolean, or year, or a group of further nodes. A node can be required, can have a list of options, and can repeat between a minimum and maximum count. A visibility condition hides a node until another field has a certain answer. SAILOR validates every answer against the schema, and it keeps hidden answers instead of deleting them. When reviewers rename a node, SAILOR carries the existing answers over.

c) Independent reviewers and explicit consolidation: Each reviewer works in their own *seat*. A project has between one and ten reviewer seats and one additional consolidation seat. Reviewers do not see each other's answers while they extract. The *consolidator* then compares the seats and decides on the final answer. In this way, the project keeps the individual answers and the consolidated result side by side.

d) A recorded history: SAILOR uses git to record and share every change. Ram [30] argues that git makes scientific work more transparent and reproducible. We follow this argument for review data. A commit shows who changed which answer and when, and a merge combines the work of reviewers who worked in parallel.

e) The human decides: SAILOR proposes and the reviewer decides. This principle applies to duplicates, to screening, and to LLM assistance. For example, SAILOR never merges a probable duplicate on its own, and it never writes an LLM answer into the project before the reviewer applies it. LLM support is off by default, and each project must enable it.

C. Running Example

The running example consists of two sample projects in the repository. The screening project contains eight candidate papers on code search and program repair. It defines eight exclusion reasons, for example, "Wrong topic", "Not peer-reviewed", and "Duplicate". In the example, the screening result includes five papers and excludes three. SAILOR then creates the extraction project "Example SLR: Code Search and Program Repair" from the screening result. The extraction project carries over the five included papers, and it records that it was created from a screening project. Its schema asks whether a paper is relevant, which study type it reports, and in which year it appeared. A repeatable group *Findings* holds a claim, the evidence for the claim with metric and evaluation type, and a confidence value. A second group *Threats to Validity* repeats at most three times and records the kind of each threat and whether the authors addressed it. Both projects have two reviewer seats and allow LLM assistance. We use this example in the walkthrough in the next section.

IV. SAILOR FROM THE REVIEWER'S VIEW

In this section, we walk through SAILOR along the activities of an SLR. We use the running example from Section III-C.

A. Planning: Protocol and Schema

Reviewers start in the project editor. There, they set the project title, the number of reviewer seats, and whether the project is a screening project. An optional review protocol records the research questions, search strings, databases,

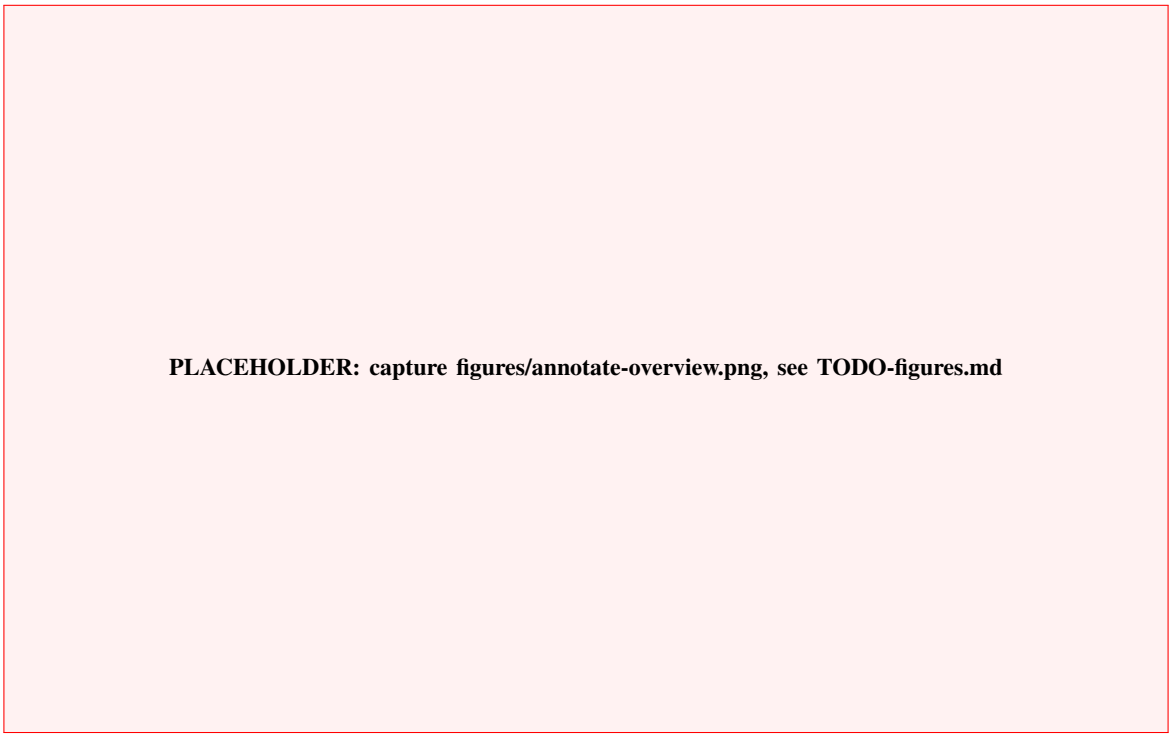


Fig. 1. Main view of SAILOR with the paper list on the left, the PDF in the middle, and the extraction form of the running example for reviewer seat 1 on the right

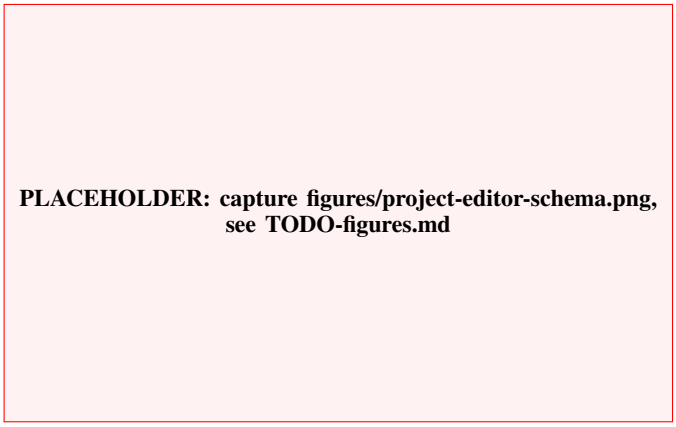


Fig. 2. Schema editor with field types, counts, options, and visibility conditions

search date, and further notes. The main part of the editor is the schema, as shown in Figure 2. Each row defines one node with its name, type, minimum and maximum count, options, description, and visibility condition. In the figure, the field *Exclusion Reason* is only visible if the field *Relevant* has the answer "No". Reviewers can reorder and nest rows by drag and drop.

B. Conducting: Import and Screening

Reviewers import references from BibTeX, RIS, or CSL-JSON files, and they can import a folder of PDFs including its subfolders. During import, SAILOR sorts each record into one of three classes: new, certain duplicate, or probable duplicate. A certain duplicate has the same normalized DOI or title as a paper in the project. A probable duplicate has a fuzzy title similarity of at least 0.90, or the same base title and an author similarity of at least 0.50. The reviewer must decide on each probable duplicate. If the reviewer merges two records, SAILOR only fills fields that are empty in the existing paper.

In a screening project, each reviewer decides per paper whether to include it, to exclude it with a reason, or to leave it undecided. Figure 3 shows the screening view of the running example. The view shows the abstract instead of the PDF, because reviewers usually screen titles and abstracts first. If the abstract was extracted from the PDF and not imported, SAILOR shows a warning, because the extraction can be wrong. Reviewers can screen with the keyboard: `I` includes, `E` excludes, `U` clears the decision, and the number keys exclude with the matching reason. After a decision, SAILOR moves on to the next paper. A progress line shows how many papers the reviewer screened, included, and excluded. The screening summary lists these counts with one row per exclusion reason, which is the input for a PRISMA flow diagram [4]. Validation reports excluded papers without a reason.

When screening is done, reviewers create the extraction project from the screening project. SAILOR carries over all included papers, never carries over excluded papers, and lets the reviewer choose whether to carry over undecided papers. The new project records where it came from and how many papers were included, excluded, and carried over.

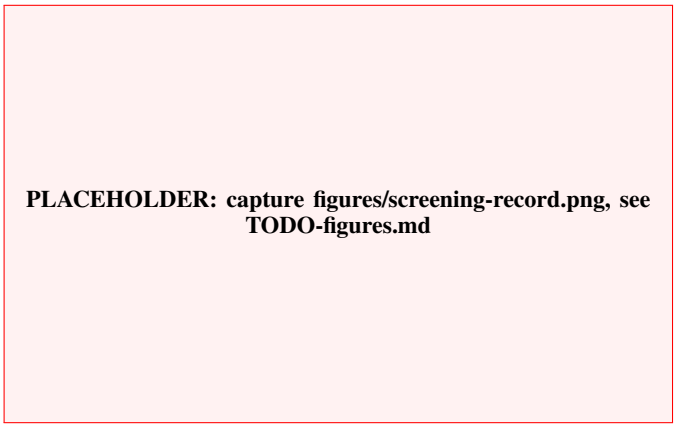


Fig. 3. Screening view with the abstract, the decision buttons, and the progress line

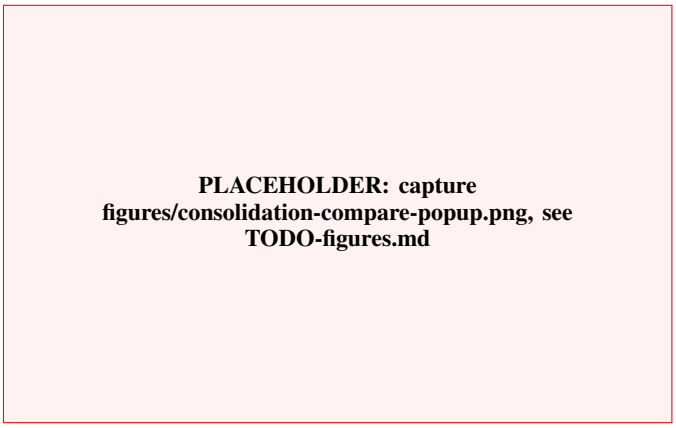


Fig. 4. Comparison of two reviewers' answers in the consolidation seat

C. Conducting: Data Extraction

Figure 1 shows the main view during extraction. The left pane lists the papers, the middle pane shows the PDF, and the right pane shows the form that SAILOR generates from the schema. Repeatable groups such as *Findings* get buttons to add and remove entries. A dot next to each paper shows how complete the paper is for the current seat. Reviewers mark a paper as finished explicitly. Thus, "finished" is a decision of the reviewer and not just the absence of empty fields.

Reviewers can copy selected text from the PDF into a field. They can also highlight passages in five colors and add notes. Each reviewer seat has its own marks. A reviewer can link a mark to one or more fields as evidence. The link popover of a field lists the linked marks, and a click on a mark jumps to the passage in the PDF. In this way, the project stores where in the paper an answer comes from. Reviewers can export their marks as standard PDF annotations into a new PDF file.

D. Conducting: Consolidation

In a project with several reviewers, every reviewer must pick a seat before editing. A reviewer only sees the answers of their own seat. The consolidator works in the additional consolidation seat.

Before SAILOR can compare answers, it must find out which entries of a repeatable group belong together. For example, two reviewers can list the findings of a paper in different orders. SAILOR matches the entries with a maximum-weight assignment over their pairwise similarity, and it keeps entries with a similarity of 0.5 or lower apart. The alignment is deterministic, so that all reviewers see the same result. Before SAILOR compares two answers, it trims them, converts them to lowercase, and collapses whitespace. If at least two reviewers answered a field and all answers are equal, SAILOR fills the consolidated value automatically and marks it as auto-filled.

For the remaining fields, the consolidator opens a comparison, as shown in Figure 4. The comparison lists the answers of all reviewers. The consolidator adopts one answer or marks the answers as equivalent. The disagreements of a paper or of the whole project can be exported as text, for example, to discuss them in a meeting.

SAILOR also computes the inter-rater agreement. For exactly two reviewers, it computes Cohen's κ [31], and for two or more reviewers Fleiss' κ [32]. In both cases, it also computes Krippendorff's α for nominal data [33], which tolerates missing answers. SAILOR shows the agreement overall and per field. If a statistic is not defined, for example, because all reviewers gave the same answer to every item, SAILOR reports the case instead of a number. In a screening project, the agreement only considers the decision.

E. Collaboration with Git

SAILOR has a built-in git view that uses the git installation of the user. Instead of a line diff of JSON files, the view lists each changed field with its previous and current value. Reviewers select which field changes they commit. A pull fetches the upstream branch and reports whether the project is up to date, can be fast-forwarded, or needs a merge. SAILOR only pushes when the reviewer asks for it. It also supports branches, a project-scoped history, and stashes of the project's own files.

For a merge, SAILOR does not use the line-based merge of git. It runs a three-way merge on the level of fields. If only one side changed a field, the merge keeps that change. The schema is merged node by node, and answers under a removed field are kept. Only if both sides changed the same field to different values does the reviewer have to decide. Figure 5 shows the dialog for these conflicts. If a merge leaves conflicts in files that do not belong to the project, SAILOR aborts it and restores the previous state.

F. LLM Assistance

LLM assistance is off unless the project enables it. Reviewers can connect nine provider application programming interfaces (APIs), among them Anthropic, OpenAI, Google Gemini, and any OpenAI-compatible server. Local models run, for example, in Ollama or a llama.cpp server [34]. SAILOR stores API keys in the keychain of the operating system. Every

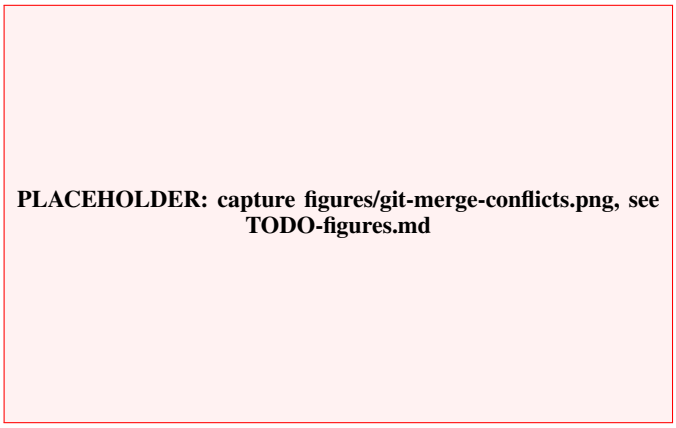


Fig. 5. Field-level resolution of merge conflicts after a pull

LLM window shows a notice that the feature is a preview and that each proposal must be checked.

For extraction, the model reads the paper and proposes answers. SAILOR validates every proposal against the schema. A review table shows each proposal with the field, the value, a quote from the paper, and a confidence. In agent mode, the model can search and read pages of the paper, query OpenAlex and Crossref, and fetch web pages. SAILOR checks whether the quoted evidence occurs in the paper, and an optional judge model checks the proposals in a bounded loop. Nothing changes in the project until the reviewer applies the selected rows. Applying is one undo step and does not overwrite existing answers. SAILOR marks each field that the model wrote until the reviewer interacts with the field. For each apply, SAILOR records the provider, the model, and the time in the project, and the paper shows a disclosure line such as "AI-assisted (2×)".

In a project with several reviewers, the last seat can be dedicated to the model. The answers of this seat go through the same consolidation as the answers of the human reviewers, so that the agreement statistics also cover the model. For screening, the model proposes a decision and a reason per paper. SAILOR never pre-selects an exclusion, and it pre-selects an inclusion only if the confidence is at least 0.8.

V. Architecture Decisions

In this section, we describe the main architecture decisions behind SAILOR. For each decision, we state the context, the decision, and its consequences.

A. Overview

SAILOR is a desktop application built with Tauri 2 [35]. The user interface is a React application in TypeScript that runs in the web view of the operating system. It renders PDFs with pdf.js and keeps the project state in a single store. A backend in Rust provides the native capabilities, such as file access, git, the keychain, and network requests. The user interface reaches the backend only through commands that the backend registers. At the time of writing, the TypeScript code base has about 50,000 lines of code without tests and about 44,000 lines of tests. The Rust backend has about 11,500 lines in 17 modules.

B. Local Files Instead of a Server

Hosted tools make collaboration easy, but they keep the review data on a server (cf. Section II-B). For a research tool, this setup has drawbacks. The data can be hard to export completely, the service can change or end, and reviewers might not be allowed to upload papers. Therefore, we decided to keep all data in local files, following the idea of local-first software [29]. The project file and the annotation files are JSON files that people can read. SAILOR keeps unknown keys in the project file, so that an older version does not delete data of a newer version. Each annotation file names the schema version it was written under, and validation reports files with an unknown schema version. The serialization is deterministic, and opening a project writes nothing. As a consequence, a version control system only sees changes that a reviewer made. The price of this decision is that SAILOR needs its own collaboration mechanism.

C. Git as Collaboration Layer

We decided to use git for collaboration instead of building a synchronization server. Git is common in SE research, offers hosting at many institutions, and keeps a full history [30]. To keep merges small, SAILOR splits the annotations into one file per paper and seat. Two reviewers who work on different seats never edit the same annotation file.

However, a line-based merge does not understand JSON. When two people edit the same file, for example, the consolidator and a reviewer who changes the schema, git can report conflicts in lines that do not conflict in their meaning. Therefore, SAILOR runs its own three-way merge on the parsed files. It compares both sides to the common base field by field. A side that did not change a value does not influence the result. Repeatable entries, project settings, and schema nodes are merged separately, and answers under a removed schema field are kept.

Using the git installation of the user brings risks. A cloned repository can contain a configuration that runs code, for example, hooks or a file system monitor. SAILOR runs git only in the backend and overrides these settings on every call, including the pager, the editor, and the `ext` protocol. It disables interactive prompts, limits the run time of each command, and validates clone URLs, paths, and branch names. Git operations are only allowed on repositories that the user opened or cloned in the current session. Before SAILOR commits or discards selected field changes, it checks that the file on disk still matches the reviewed state, and it refuses the write otherwise.

D. From Electron to Tauri

Up to version 1.x, SAILOR was an Electron application [36]. The Electron main process had about 5,600 lines of TypeScript and 98 inter-process communication (IPC)

handlers. Electron bundles its own Chromium and Node.js runtime, and the main process could use the full Node.js API.

For version 2.0.0, we moved SAILOR to Tauri 2 [35]. Tauri uses the web view of the operating system and a backend in Rust. Tauri moves the security boundary into Rust and offers a capability system and a signed updater. We split the responsibilities as follows. Rust owns everything that needs authorization: file input and output, process spawning, downloads, and access to the keychain. TypeScript keeps the parsing of files, the merge logic, and the assembly of exported PDFs, because this code was already tested and does not need native access. SAILOR does not expose the generic file system and shell plugins of Tauri to the user interface. The capability file grants the web view only the default core permissions.

The move has consequences for users and for us. SAILOR now depends on three different web views: WKWebView on macOS, WebView2 on Windows, and WebKitGTK on Linux. We checked that pdf.js renders on all three. The minimum macOS version is 15. Users of version 1.x have to install version 2.0.0 manually and enter their API keys again. Also, the guard against reload shortcuts is not ported yet (cf. Section VII).

E. Security Model

A review tool opens files and repositories from other people, and it can send data to LLM providers. Therefore, we treat the user interface as untrusted and enforce all checks in the backend. SAILOR only saves a project to a path that the user opened or chose in a native dialog during the current session. It serves a PDF only if the file lies inside the project folder. For a PDF outside the folder, SAILOR asks the user and allows only this exact path for the current session. Exports only write to destinations that the user picked in a save dialog, and SAILOR refuses symbolic links as targets. Updates are signed, and SAILOR downloads and installs them only when the user asks for it.

For LLM providers, SAILOR stores API keys in the keychain and sends a key only to the origin of its provider. It refuses redirects for requests that carry a key. The agent mode can fetch web pages, which opens the door to server-side request forgery (SSRF) [37]. SAILOR only allows `http` and `https` URLs, resolves the host name, and refuses requests to loopback, private, and other non-public addresses. It checks every redirect again, follows at most five redirects, stops after 20 seconds, reads at most 2 MB, and only accepts text content. The fetched text is marked as untrusted data, and the system prompt tells the model to never follow instructions from it.

F. Controlled LLM Assistance

Prior studies found that LLMs can help with screening and extraction but still make errors [6], [7], [24]. Therefore, we designed the LLM support so that it cannot change the review data without a human decision. Suggestions live outside the project until the reviewer applies them. SAILOR validates them against the schema and checks quoted evidence against the text of the paper. The durable record of each apply makes the use of LLMs visible in the data, so that authors can report it. Treating the model as an additional reviewer seat reuses the consolidation and agreement statistics, instead of adding a separate evaluation mechanism.

Some reviewers cannot send papers to a hosted provider. For them, SAILOR supports local models. For chat models, users connect a local server such as Ollama, LM Studio, llama.cpp, or vLLM. For small decision models, SAILOR can manage a llama.cpp server itself [34]. It downloads the server and the models only from an allowlist of sources, verifies their SHA-256 checksums, and starts the server on `127.0.0.1` with a random key per session. The server uses a graphics processing unit (GPU) when one is available and the central processing unit (CPU) otherwise. We verified this runtime on macOS on arm64. The managed installation needs a llama.cpp release with a specific route for decision models. As long as no release includes this route, the setup dialog keeps the installation disabled and points users to a server that they run themselves.

VI. Requirements and Traceability

SAILOR asks reviewers to keep their decisions traceable. We think that the tool should meet the same standard for its own behavior. Traceability links requirements to the artifacts that realize and verify them [38], [39]. In this section, we describe how we documented the requirements of SAILOR and how far they are traced.

A. Recovery of the Requirements

SAILOR grew without a written specification. Therefore, we reverse-engineered the requirements from the implementation of version 1.8.1 in August 2026. We updated them in September 2026 and for the move to Tauri in version 2.0.0. Every requirement describes behavior that the code implements, and every link is a recovered link and not one that we created during development.

We used four recovery techniques, in descending order of evidential weight. First, we read the names and assertions of tests as executable specifications. For example, a store test named “clears the reason in the same undo step” pins the requirement that a paper loses its exclusion reason in the same undo step when its decision changes from “exclude” to another state. Second, we inspected the source code and recorded the implementing files and lines. Third, we searched the commit history for the commits that introduced or pinned a behavior. Fourth, we mapped each requirement to the developer wiki, the schema reference, and the user guide.

Each requirement has an identifier with a category prefix, for example, `REQ-CON-80` for consolidation. The identifiers advance in steps of ten, so that new requirements fit in without renumbering. An entry consists of a title, a description in the form “The system shall …”, a type, the evidence, a status, and the tests. The type states whether the requirement is functional or non-functional and names its quality characteristic according to ISO/IEC 25010 [40]. The requirements are split into seven files by category. A separate traceability matrix lists one

TABLE II
REQUIREMENTS PER CATEGORY AND TRACE LINKS OF ALL 389 REQUIREMENTS

Category	Requirements	
Data model and project editor	61	
Annotation view, paper list, and PDF	54	
Screening	39	
Consolidation	44	
Git integration	59	
LLM assistance	88	
Platform and desktop shell	46	
Trace link	**Count**	**Share**
Implementing source code	365	93%
At least one verifying test	360	92%
Source inspection only	31	8%
Introducing or pinning commit	118	30%
Documentation section	337	86%

row per requirement with the columns *Implemented in*, *Verified by*, *History*, and *Docs*. An empty *Verified by* cell means that we confirmed the requirement only by inspecting the source code.

B. Coverage

In total, we documented 389 requirements. Table II shows how they are distributed over the categories and how many of them are linked to each kind of artifact. Of the requirements, 343 are functional and 46 are non-functional, and 32 have the type security. At the time of writing, 390 requirements are implemented. One requirement is implemented in part: the guard against reload shortcuts (cf. Section VII). The links point to 177 distinct source files and 185 distinct test files. Of all requirements, 93% link to implementing source code, and 92% link to at least one verifying test. The remaining 31 requirements are confirmed by source inspection only. Of these, 15 belong to the desktop shell and 12 to the git integration. Only 30% link to a commit that introduced or pinned the behavior.

We count these numbers with a small script in the report folder that parses the requirement files and the matrix. The script also checks that every linked file exists.

C. Automated Tests

SAILOR has four test suites. The unit suite has 2834 tests in 197 files and runs with Vitest in jsdom, a simulated browser environment. The integration suite has 56 tests in 11 files. These tests render the real components and run scenarios such as annotation, consolidation, and merges against scratch git repositories. The Rust backend has 104 tests. Finally, a probe build of the desktop application opens the sample project, renders a PDF, and checks that a path traversal is refused. The continuous integration (CI) runs this probe in the system web view on macOS, Windows, and Linux. On Linux and Windows, it also drives the probe through WebDriver.

To keep the links current, the repository asks every change that affects a requirement to update the matching requirement file in the same change. This rule is a convention and not a check, so the links can still drift.

VII. DISCUSSION AND LIMITATIONS

In this section, we discuss the limitations of SAILOR and of this report. For each limitation, we state what it affects and how we mitigate it.

A. Limitations

We have not evaluated SAILOR with users. This report describes the tool and its design, but it does not show that SAILOR makes reviews faster, more accurate, or easier to trace than other tools. We derived the design from the guidelines for SLRs, and it might not match the needs of every research group. A user study is needed to assess the benefit (cf. Section VIII).

The requirements were reverse-engineered from the code by the developer of the tool. Thus, they describe what SAILOR does and not what users need. If the code has a bug, a requirement can describe the buggy behavior as intended. Also, the developer might have overlooked behavior or described it in favor of the implementation. We mitigate this threat with the evidence of each requirement, because tests, source lines, and commits can be checked by others. Still, an independent review of the requirements is open.

The move to Tauri is recent. Six features of the Electron shell are now ported to Tauri: the single application instance, the persisted window state, the migration of the old settings folder, the denial of device permissions, the opening of external links in the system browser, and the undo entry in the application menu. The guard against reload shortcuts is not ported yet. The renderer registers a guard while changes are unsaved, but the Tauri shell has no reload confirmation dialog of its own, and its View menu has no reload item. Furthermore, all figures in this report show the user interface of version 1.x.

Our tests cover the three operating systems unevenly. The probe of the desktop application runs on macOS, Windows, and Linux, but a WebDriver session drives it only on Linux and Windows, because the Tauri driver does not support macOS. Most behavior is tested in the unit and integration suites, which run in a simulated browser and not in the web views of the operating systems. Differences between WKWebView, WebView2, and WebKitGTK can therefore remain unnoticed. We checked the rendering of PDFs on all three web views, but the probe does not yet cover further workflows.

The installers are not signed with a certificate from Apple or Microsoft. macOS and Windows warn users the first time they open SAILOR, and users have to confirm the start. This step can confuse users and can also train them to ignore such warnings. The README explains the steps for each operating system. Updates are signed with a key of the project, and SAILOR verifies the signature before it installs an update.

We verified the managed local runtime for decision models only on macOS on arm64. The code paths for GPUs with CUDA and Vulkan have not run on real hardware. In addition, the managed installation stays disabled until llama.cpp releases

the route that the decision models need. Until then, users who want local decision models have to run a server themselves. Hosted providers and local chat servers are not affected by this limitation.

The LLM assistance is a preview. We have not measured how accurate its proposals are, and studies report errors of LLMs in screening and extraction [7], [24]. A reviewer who accepts proposals without checking them can bring wrong data into the review. SAILOR reduces this risk, because it checks the quoted evidence, marks the fields that the model wrote, and records each use for the report. It cannot replace the judgment of the reviewer.

SAILOR does not cover all activities of an SLR. It records the search strings and databases, but it does not run searches. It also does not analyze or synthesize the extracted data. The extracted data stays in JSON files, so reviewers need their own scripts or tools for the analysis.

B. Availability

SAILOR is open source under the GPL-3.0 license. The source code is available on GitHub.[1] Version 2.0.0 provides installers for macOS 15 or later on Apple silicon and Intel, for Windows, and for Linux as AppImage, Debian, and RPM packages. The release workflow builds the installers from a version tag and archives each release on Zenodo [15]. The repository also contains a sample screening project and a sample extraction project, a user guide, and the requirement files described in Section VI.

VIII. Conclusion

In this paper, we looked into tool support that keeps the data of a systematic literature review (SLR) traceable. We presented SAILOR, an open-source desktop tool for screening, data extraction, and consolidation. A SAILOR project consisted of plain JSON files that a schema described and that git versioned. Reviewers worked independently in their own seats and linked their answers to evidence in the papers. A consolidator resolved disagreements with the help of agreement statistics, and SAILOR merged concurrent changes field by field. Optional large language model (LLM) assistance proposed answers with evidence, but a human reviewed every proposal before SAILOR applied it.

We also explained the main architecture decisions. These were the local files instead of a server, git as collaboration layer, the security model, and the move from Electron to Tauri in version 2.0.0. To keep the tool itself traceable, we documented 389 requirements. Of these requirements, 93% linked to implementing source code and 92% linked to at least one automated test. The test suites held 2834 unit tests, 56 integration tests, and 104 Rust tests. We released SAILOR 2.0.0 under the GPL-3.0 license and archived it on Zenodo.

Regarding the tool, we want to port the guard against reload shortcuts to Tauri and retest the user interface in the web views of all three operating systems. We also want to verify the local model runtime on graphics processing units (GPUs) with CUDA and Vulkan and enable the managed installation once llama.cpp provides the required route.

In future research, we want to evaluate SAILOR in a user study with reviewers who conduct a real SLR. We want to find out whether the tool reduces the effort of extraction and consolidation and whether the recorded history helps others to trace the review. We also want to measure how accurate the LLM proposals are and how they influence the decisions of reviewers.

Lastly, we want to have the requirements reviewed by people other than the developer and check the trace links automatically in the continuous integration (CI).

[1] https://github.com/Gram21/SaiLoR

References

[1] B. Kitchenham and S. Charters, "Guidelines for performing systematic literature reviews in software engineering," Keele University and Durham University, Tech. Rep. EBSE-2007-01, 2007.

[2] K. Petersen, S. Vakkalanka, and L. Kuzniarz, "Guidelines for conducting systematic mapping studies in software engineering: An update," *Information and Software Technology*, vol. 64, pp. 1–18, 2015.

[3] C. Wohlin, "Guidelines for snowballing in systematic literature studies and a replication in software engineering," in *Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering (EASE)*. ACM, 2014, pp. 1–10.

[4] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, L. Shamseer, J. M. Tetzlaff, E. A. Akl, S. E. Brennan, R. Chou, J. Glanville, J. M. Grimshaw, A. Hróbjartsson, M. M. Lalu, T. Li, E. W. Loder, E. Mayo-Wilson, S. McDonald, L. A. McGuinness, L. A. Stewart, J. Thomas, A. C. Tricco, V. A. Welch, P. Whiting, and D. Moher, "The PRISMA 2020 statement: an updated guideline for reporting systematic reviews," *BMJ*, vol. 372, p. n71, 2021.

[5] E. Hassler, J. C. Carver, N. A. Kraft, and D. Hale, "Outcomes of a community workshop to identify and rank barriers to the systematic literature review process," in *Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering (EASE)*. ACM, 2014, pp. 1–10.

[6] K. R. Felizardo, M. S. Lima, A. Deizepe, T. U. Conte, and I. Steinmacher, "ChatGPT application in systematic literature reviews in software engineering: an evaluation of its accuracy to support the selection activity," in *Proceedings of the 18th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM)*. ACM, 2024, pp. 25–36.

[7] Q. Khraisha, S. Put, J. Kappenberg, A. Warraitch, and K. Hadfield, "Can large language models replace humans in systematic reviews? Evaluating GPT-4's efficacy in screening and extracting data from peer-reviewed and grey literature in multiple languages," *Research Synthesis Methods*, vol. 15, no. 4, pp. 616–626, 2024.

[8] C. Marshall and P. Brereton, "Tools to support systematic literature reviews in software engineering: A mapping study," in *ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM)*. IEEE, 2013, pp. 296–299.

[9] C. Marshall, P. Brereton, and B. Kitchenham, "Tools to support systematic reviews in software engineering: a feature analysis," in *Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering (EASE)*. ACM, 2014, pp. 1–10.

[10] H. Harrison, S. J. Griffin, I. Kuhn, and J. A. Usher-Smith, "Software tools to support title and abstract screening for systematic reviews in healthcare: an evaluation," *BMC Medical Research Methodology*, vol. 20, p. 7, 2020.

[11] Veritas Health Innovation, "Covidence systematic review software," Melbourne, Australia. https://www.covidence.org, accessed 2026-11-01.

[12] M. Ouzzani, H. Hammady, Z. Fedorowicz, and A. Elmagarmid, "Rayyan—a web and mobile app for systematic reviews," *Systematic Reviews*, vol. 5, no. 1, p. 210, 2016.

[13] C. Kohl, E. J. McIntosh, S. Unger, N. R. Haddaway, S. Kecke, J. Schiemann, and R. Wilhelm, "Online tools supporting the conduct and reporting of systematic reviews and systematic maps: a case study on CADIMA and review of existing tools," *Environmental Evidence*, vol. 7, no. 1, p. 8, 2018.

[14] R. van de Schoot, J. de Bruin, R. Schram, P. Zahedi, J. de Boer, F. Weijdema, B. Kramer, M. Huijts, M. Hoogerwerf, G. Ferdinands, A. Harkema, J. Willemsen, Y. Ma, Q. Fang, S. Hindriks, L. Tummers, and D. L. Oberski, "An open source machine learning framework for efficient and transparent systematic reviews," *Nature Machine Intelligence*, vol. 3, no. 2, pp. 125–133, 2021.

[15] J. Keim, "SaiLoR 2.0.0," Software. Source code: https://github.com/Gram21/SaiLoR, 2026, zenodo concept DOI, always resolves to the latest version.

[16] B. A. Kitchenham, D. Budgen, and P. Brereton, *Evidence-Based Software Engineering and Systematic Reviews*. Chapman and Hall/CRC, 2015.

[17] A. Al-Zubidy and J. C. Carver, "Identification and prioritization of SLR search tool requirements: an SLR and a survey," *Empirical Software Engineering*, vol. 24, no. 1, pp. 139–169, 2019.

[18] K. R. Felizardo and J. C. Carver, "Automating systematic literature review," in *Contemporary Empirical Methods in Software Engineering*, M. Felderer and G. H. Travassos, Eds. Springer, 2020, pp. 327–355.

[19] R. van Dinter, B. Tekinerdogan, and C. Catal, "Automation of systematic literature reviews: A systematic literature review," *Information and Software Technology*, vol. 136, p. 106589, 2021.

[20] V. Freitas, "Parsifal," Software. https://parsif.al, source code at https://github.com/vitorfs/parsifal, accessed 2026-11-02.

[21] E. Hernandes, A. Zamboni, S. Fabbri, and A. Di Thommazo, "Using GQM and TAM to evaluate StArt – a tool that supports systematic review," *CLEI Electronic Journal*, vol. 15, no. 1, 2012.

[22] S. Fabbri, C. Silva, E. Hernandes, F. Octaviano, A. Di Thommazo, and A. Belgamo, "Improvements in the StArt tool to better support the systematic review process," in *Proceedings of the 20th International Conference on Evaluation and Assessment in Software Engineering (EASE)*. ACM, 2016, pp. 1–5.

[23] A. M. Fernández-Sáez, M. Genero Bocco, and F. P. Romero, "SLR-TOOL - a tool for performing systematic literature reviews," in *Proceedings of the 5th International Conference on Software and Data Technologies (ICSOFT)*. SciTePress, 2010, pp. 157–166.

[24] E. Syriani, I. David, and G. Kumar, "Screening articles for systematic reviews with ChatGPT," *Journal of Computer Languages*, vol. 80, p. 101287, 2024.

[25] K. R. Felizardo, I. Steinmacher, M. S. Lima, A. Deizepe, T. U. Conte, and M. P. Barcellos, "Data extraction for systematic mapping study using a large language model - a proof-of-concept study in software engineering," in *Proceedings of the 18th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM)*. ACM, 2024, pp. 407–413.

[26] G. Gartlehner, L. Kahwati, R. Hilscher, I. Thomas, S. Kugley, K. Crotty, M. Viswanathan, B. Nussbaumer-Streit, G. Booth, N. Erskine, A. Konet, and R. Chew, "Data extraction for evidence synthesis using a large language model: A proof-of-concept study," *Research Synthesis Methods*, vol. 15, no. 4, pp. 576–589, 2024.

[27] S. Wang, H. Scells, B. Koopman, and G. Zuccon, "Can ChatGPT write a good Boolean query for systematic review literature search?" in *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*. ACM, 2023, pp. 1426–1436.

[28] A. Alshami, M. Elsayed, E. Ali, A. E. E. Eltoukhy, and T. Zayed, "Harnessing the power of ChatGPT for automating systematic review process: Methodology, case study, limitations, and future directions," *Systems*, vol. 11, no. 7, p. 351, 2023.

[29] M. Kleppmann, A. Wiggins, P. van Hardenberg, and M. McGranaghan, "Local-first software: you own your data, in spite of the cloud," in *Proceedings of the 2019 ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software (Onward!)*. ACM, 2019, pp. 154–178.

[30] K. Ram, "Git can facilitate greater reproducibility and increased transparency in science," *Source Code for Biology and Medicine*, vol. 8, no. 1, p. 7, 2013.

[31] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960.

[32] J. L. Fleiss, "Measuring nominal scale agreement among many raters," *Psychological Bulletin*, vol. 76, no. 5, pp. 378–382, 1971.

[33] K. Krippendorff, *Content Analysis: An Introduction to Its Methodology*, 4th ed. SAGE Publications, 2019.

[34] ggml-org, "llama.cpp: LLM inference in C/C++," https://github.com/ggml-org/llama.cpp, accessed 2026-11-02.

[35] Tauri Contributors, "Tauri 2.0," https://v2.tauri.app/, accessed 2026-11-02.

[36] OpenJS Foundation, "Electron: Build cross-platform desktop apps with JavaScript, HTML, and CSS," https://www.electronjs.org/, accessed 2026-11-02.

[37] OWASP Foundation, "Server side request forgery," https://community.owasp.org/attacks/Server_Side_Request_Forgery, accessed 2026-11-02.

[38] O. C. Z. Gotel and A. C. W. Finkelstein, "An analysis of the requirements traceability problem," in *Proceedings of the IEEE International Conference on Requirements Engineering (ICRE)*. IEEE, 1994, pp. 94–101.

[39] J. Cleland-Huang, O. C. Z. Gotel, J. Huffman Hayes, P. Mäder, and A. Zisman, "Software traceability: trends and future directions," in *Future of Software Engineering Proceedings (FOSE)*. ACM, 2014, pp. 55–69.

[40] ISO/IEC, "ISO/IEC 25010:2011 systems and software engineering – systems and software quality requirements and evaluation (SQuaRE) – system and software quality models," International Organization for Standardization, 2011.